



DC – AI – Routing – Switching – Security – Wireless – Automation

Hybrid MNHA with eBGP

An HPE Juniper Networking Techpost

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HPE



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1 Hybrid MNHA with eBGP

James Rathbun - 06/12/2025

Let's highlight the flexibility of Multi-Node High Availability (MNHA) and JUNOS while providing design considerations when implementing MNHA in a hybrid deployment model.

1.1 Introduction

Every network environment is unique and has different requirements and caveats. The focus will be on:

- Implementation considerations
- Monitoring/Failover scenarios
- MNHA functionality and traffic flows (with hybrid deployments)

Steven Jacques' post, [Multi-Node High Availability](#) is an excellent post covering the basics and MNHA and Layer 3 (L3) implementations.

1.1.1 Objectives

High level topology (Figure 1) represents a topology having a Layer 2 (L2) hand-off to an external network, a Layer 3 (L3) internal hand-off demarcated by a pair of SRXs. MNHA hybrid designs provide flexibility of L2 hand-offs (internal/external). Objectives validated:

- RTRs (RTR-1 and RTR-2) send L3 untagged packets (routed) to the SRXs (SRX-1 and SRX-2)
- RTRs also send L2 tagged packet to the SRXs (SRXs will provide default gateway functionality)
- RTRs have redundant connections, crossed (bowtie) with the SRXs
- Northbound connections from the SRXs provide default gateway functionality with the external network.
- High availability (HA) implementation with fast stateful failover.

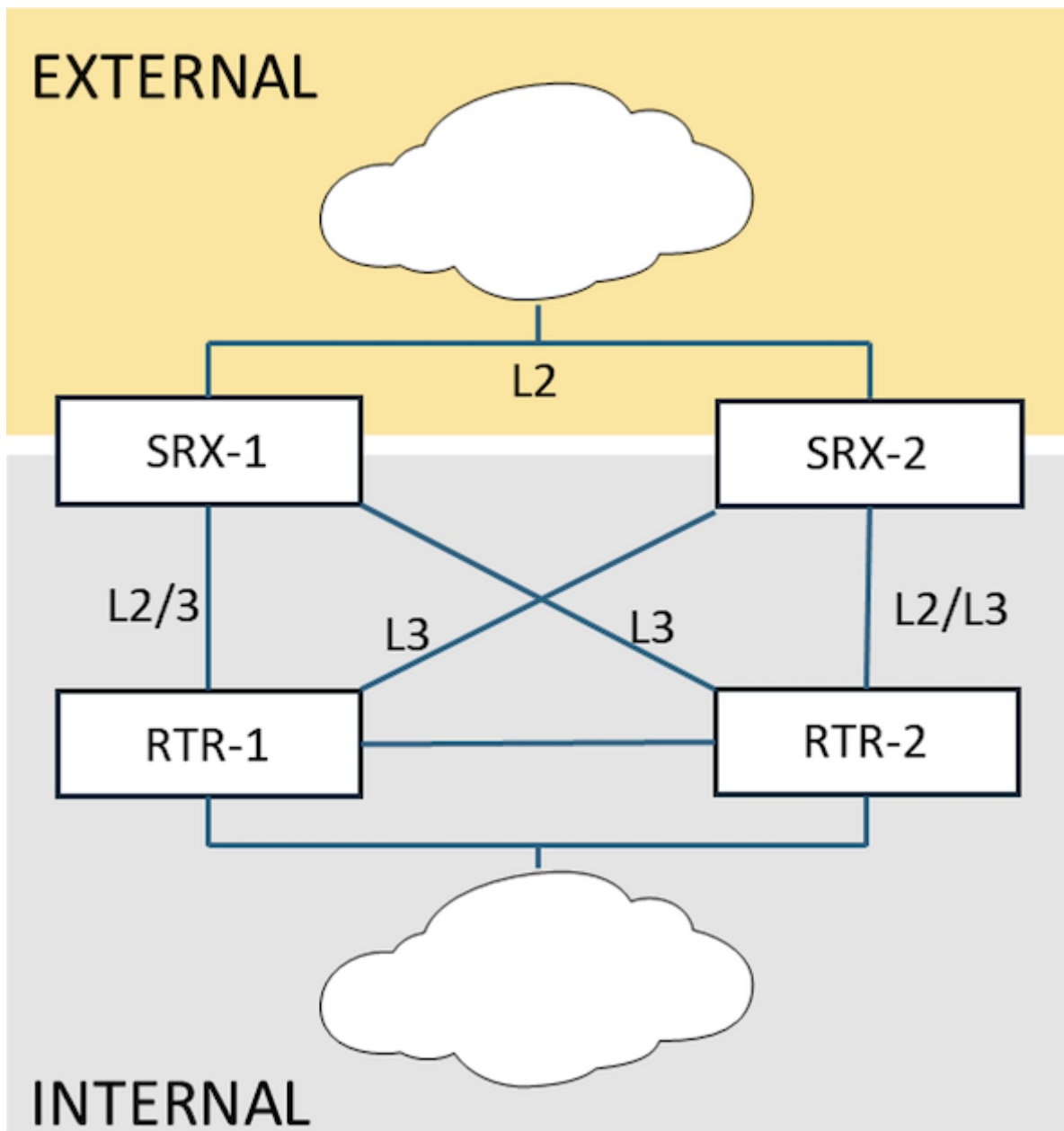
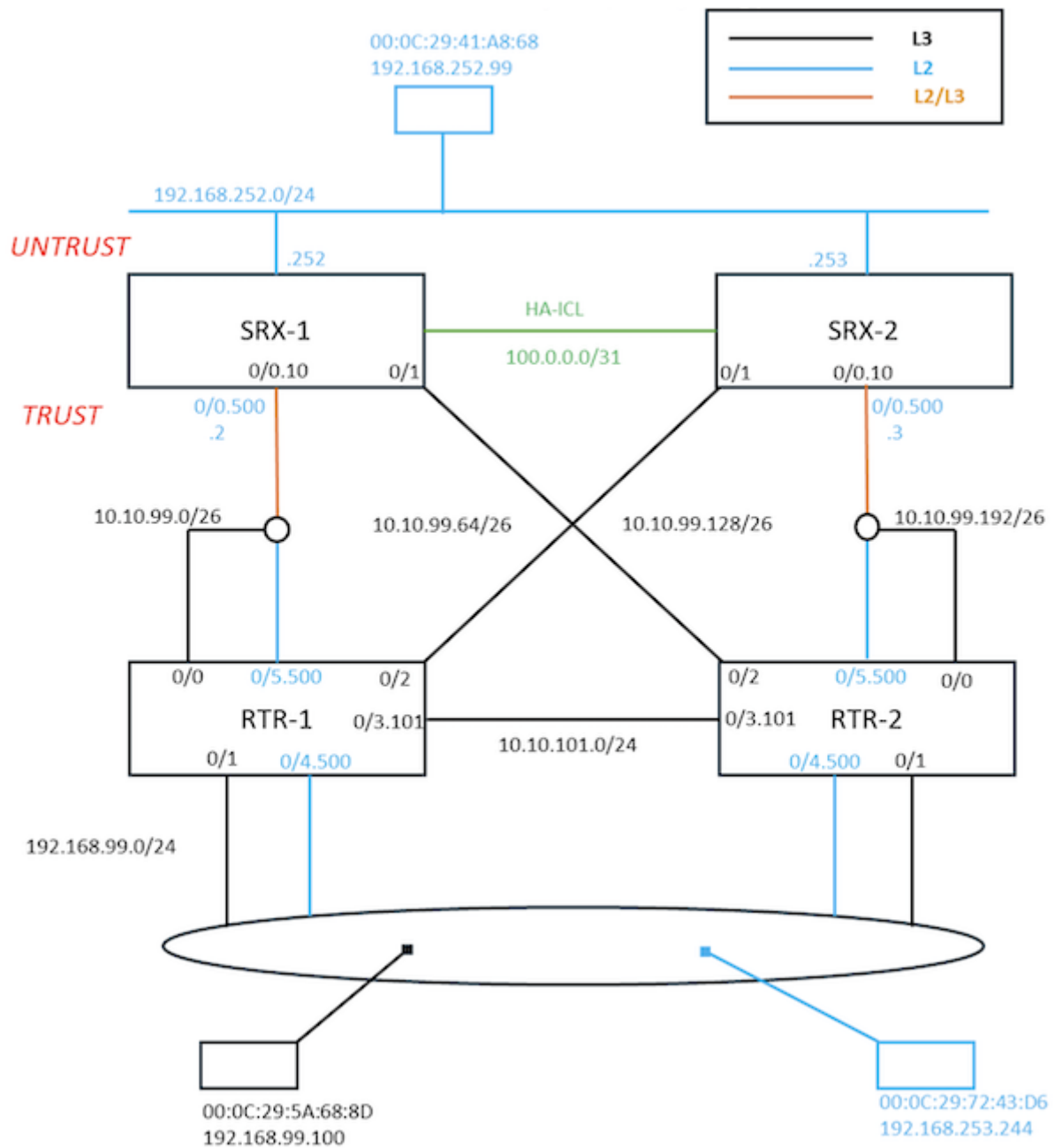


Figure 1: High Level Topology

**Figure 2:** Reference Layer 2/3 Topology

1.2 Border Gateway Protocol (BGP)

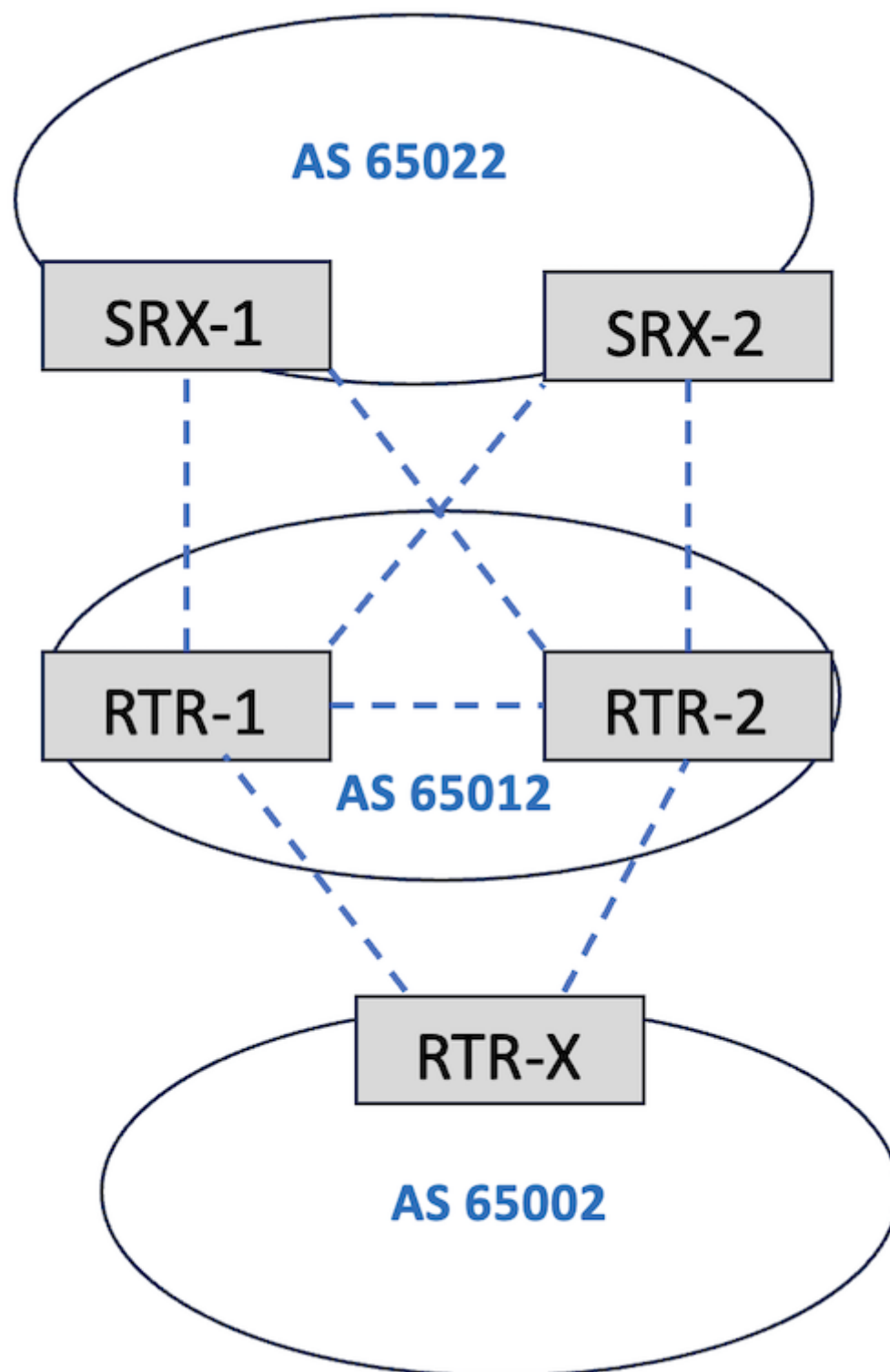


Figure 3: BGP AS Topology

```
1 MNHA-RTR-1> show bgp summary
2
3 Threading mode: BGP I/O
4 Default eBGP mode: advertise - accept, receive - accept
5 Groups: 4 Peers: 5 Down peers: 0
6 Table Tot Paths Act Paths Suppressed History Damp State Pending
7 inet.0
8          72 33 0 0 0 0
```

```

 9 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
10 10.10.99.5 65022 2155 2124 0 0 16:06:17 Establ
11   inet.0: 6/8/8/0
12 10.10.99.131 65022 2146 2122 0 0 16:05:46 Establ
13   inet.0: 2/8/8/0
14 10.10.101.2 65012 2138 2128 0 0 16:06:59 Establ
15   inet.0: 0/30/30/0
16 192.168.99.1 65002 2221 2126 0 0 16:06:54 Establ
17   inet.0: 25/26/26/0

```

```

1 MNHA-RTR-2> show bgp summary
2
3 Threading mode: BGP I/O
4 Default eBGP mode: advertise - accept, receive - accept
5 Groups: 4 Peers: 5 Down peers: 0
6 Table Tot Paths Act Paths Suppressed History Damp State Pending
7 inet.0
8           75 33 0 0 0 0
9 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
10 10.10.99.69 65022 5 4 0 0 9 Establ
11   inet.0: 3/8/8/0
12 10.10.99.195 65022 1967 1950 0 1 14:44:51 Establ
13   inet.0: 5/8/8/0
14 10.10.101.1 65012 2149 2159 0 0 16:15:59 Establ
15   inet.0: 0/33/33/0
16 10.10.201.1 65100 2145 2150 0 0 16:16:04 Establ
17   VR_ALT.inet.0: 1/1/1/0
18 192.168.99.1 65002 1541 1479 0 1 11:10:34 Establ
19   inet.0: 25/26/26/0

```

```

1 MNHA-SRX-1> show bgp summary
2
3 Threading mode: BGP I/O
4 Default eBGP mode: advertise - accept, receive - accept
5 Groups: 1 Peers: 2 Down peers: 0
6 Table Tot Paths Act Paths Suppressed History Damp State Pending
7 inet.0
8           50 50 0 0 0 0
9 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
10 10.10.99.1 65012 2146 2174 0 0 16:15:54 Establ
11   inet.0: 25/25/25/0
12 10.10.99.65 65012 6 5 0 0 46 Establ
13   inet.0: 25/25/25/0

```

```

1 MNHA-SRX-2> show bgp summary
2
3 Threading mode: BGP I/O
4 Default eBGP mode: advertise - accept, receive - accept
5 Groups: 1 Peers: 2 Down peers: 0
6 Table Tot Paths Act Paths Suppressed History Damp State Pending
7 inet.0
8           50 50 0 0 0 0
9 Peer AS InPkt OutPkt OutQ Flaps Last Up/Dwn State|#Active/Received/Accepted/Damped...
10 10.10.99.129 65012 8 6 0 0 1:22 Establ
11   inet.0: 25/25/25/0
12 10.10.99.193 65012 7 5 0 0 1:11 Establ
13   inet.0: 25/25/25/0

```

```

1 MNHA-RTR-1> show configuration protocols bgp | display set
2 set protocols bgp group EXT type external
3 set protocols bgp group EXT multihop ttl 4
4 set protocols bgp group EXT local-address 192.168.99.178
5 set protocols bgp group EXT local-as 65012
6 set protocols bgp group EXT multipath multiple-as
7 set protocols bgp group EXT bfd-liveness-detection minimum-interval 500
8 set protocols bgp group EXT bfd-liveness-detection multiplier 3
9 set protocols bgp group EXT neighbor 192.168.99.1 peer-as 65002
10 set protocols bgp group INT type internal
11 set protocols bgp group INT export NHS
12 set protocols bgp group INT local-as 65012
13 set protocols bgp group INT bfd-liveness-detection minimum-interval 500
14 set protocols bgp group INT bfd-liveness-detection minimum-receive-interval 500
15 set protocols bgp group INT bfd-liveness-detection multiplier 3
16 set protocols bgp group INT neighbor 10.10.101.2
17 set protocols bgp group INT-ebgp type external
18 set protocols bgp group INT-ebgp local-as 65012
19 set protocols bgp group INT-ebgp bfd-liveness-detection minimum-interval 500
20 set protocols bgp group INT-ebgp bfd-liveness-detection multiplier 3

```

```

21 set protocols bgp group INT-ebgp bfd-liveness-detection holddown-interval 10000
22 set protocols bgp group INT-ebgp neighbor 10.10.99.131 peer-as 65022
23 set protocols bgp group INT-ebgp neighbor 10.10.99.5 peer-as 65022

```

```

1 MNHA-RTR-2> show configuration protocols bgp | display set
2 set protocols bgp group unINT type external
3 set protocols bgp group unINT local-address 192.168.99.179
4 set protocols bgp group unINT local-as 65012
5 set protocols bgp group unINT multipath multiple-as
6 set protocols bgp group unINT bfd-liveness-detection minimum-interval 500
7 set protocols bgp group unINT bfd-liveness-detection minimum-receive-interval 500
8 set protocols bgp group unINT bfd-liveness-detection multiplier 3
9 set protocols bgp group unINT neighbor 192.168.99.1 peer-as 65002
10 set protocols bgp group INT type internal
11 set protocols bgp group INT export NHS
12 set protocols bgp group INT local-as 65012
13 set protocols bgp group INT bfd-liveness-detection minimum-interval 500
14 set protocols bgp group INT bfd-liveness-detection multiplier 3
15 set protocols bgp group INT neighbor 10.10.101.1
16 set protocols bgp group INT-ebgp local-as 65012
17 set protocols bgp group INT-ebgp bfd-liveness-detection minimum-interval 500
18 set protocols bgp group INT-ebgp bfd-liveness-detection multiplier 3
19 set protocols bgp group INT-ebgp bfd-liveness-detection holddown-interval 10000
20 set protocols bgp group INT-ebgp neighbor 10.10.99.69 peer-as 65022
21 set protocols bgp group INT-ebgp neighbor 10.10.99.195 peer-as 65022

```

```

1 MNHA-SRX-1> show configuration protocols bgp | display set
2 set protocols bgp group trust type external
3 set protocols bgp group trust export MNHA_ROUTE_POLICY
4 set protocols bgp group trust local-as 65022
5 set protocols bgp group trust multipath
6 set protocols bgp group trust bfd-liveness-detection minimum-interval 500
7 set protocols bgp group trust bfd-liveness-detection minimum-receive-interval 500
8 set protocols bgp group trust bfd-liveness-detection multiplier 3
9 set protocols bgp group trust bfd-liveness-detection holddown-interval 10000
10 set protocols bgp group trust neighbor 10.10.99.1 peer-as 65012
11 set protocols bgp group trust neighbor 10.10.99.65 peer-as 65012

```

```

1 MNHA-SRX-2> show configuration protocols bgp | display set
2 set protocols bgp group trust type external
3 set protocols bgp group trust export MNHA_ROUTE_POLICY
4 set protocols bgp group trust local-as 65022
5 set protocols bgp group trust multipath multiple-as
6 set protocols bgp group trust bfd-liveness-detection minimum-interval 500
7 set protocols bgp group trust bfd-liveness-detection multiplier 3
8 set protocols bgp group trust bfd-liveness-detection holddown-interval 10000
9 set protocols bgp group trust neighbor 10.10.99.129 peer-as 65012
10 set protocols bgp group trust neighbor 10.10.99.193 peer-as 65012

```

1.2.1 BGP Bi-Directional Liveliness Detection

```

1 MNHA-SRX-1> show bfd session
2
3 Address State Interface Time Interval Multiplier
4 10.10.99.1 (BGP) Up ge-0/0/0.10 1.500 0.500 3
5 10.10.99.65 (BGP) Up ge-0/0/1.0 1.500 0.500 3
6 100.0.0.1 (HA) Up 2.000 0.400 5
7
8 3 sessions, 5 clients
9 Cumulative transmit rate 6.5 pps, cumulative receive rate 6.5 pps

```

1.2.2 BFD BGP Hold-down

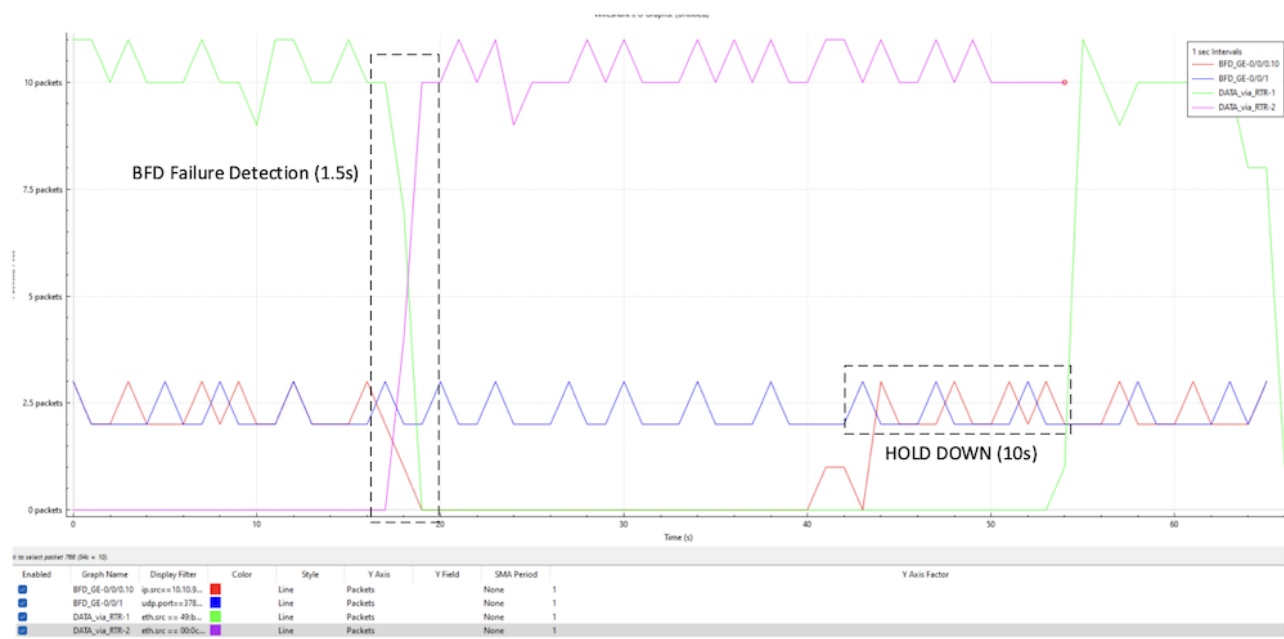


Figure 4: Figure 4. Packet Capture at SRX-1, BFD Failure Detection

1.2.3 MNHA - Activeness

The highest priority determines the active node in the MNHA pair. On SRX-1 and SRX-2:

```
1 set chassis high-availability services-redundancy-group 1 activeness-priority 200
```

(Activeness configurations)

```
1 MNHA-SRX-1> show chassis high-availability services-redundancy-group 1
2 SRG failure event codes:
3   BF BFD monitoring
4   IP IP monitoring
5   IF Interface monitoring
6   CP Control Plane monitoring
7
8 Services Redundancy Group: 1
9   Deployment Type: HYBRID
10  Status: ACTIVE
11  Activeness Priority: 200
12  Preemption: DISABLED
13  Process Packet In Backup State: NO
14  Control Plane State: READY
15  System Integrity Check: N/A
16  Failure Events: NONE
17  Peer Information:
18    Peer Id: 2
19    Status : BACKUP
20    Health Status: HEALTHY
21    Failover Readiness: READY
```

(SRX-1 Active status for SRG-1)

```
1 MNHA-SRX-2> show chassis high-availability services-redundancy-group 1
2 SRG failure event codes:
3   BF BFD monitoring
4   IP IP monitoring
5   IF Interface monitoring
6   CP Control Plane monitoring
7
8 Services Redundancy Group: 1
9   Deployment Type: HYBRID
```

```

10      Status: BACKUP
11      Activeness Priority: 100
12      Preemption: DISABLED
13      Process Packet In Backup State: NO
14      Control Plane State: READY
15      System Integrity Check: COMPLETE
16      Failure Events: NONE
17      Peer Information:
18      Peer Id: 1
19      Status : ACTIVE
20      Health Status: HEALTHY
21      Failover Readiness: N/A

```

(SRX-2 Backup status for SRG-1)

1.2.4 MNHA - Signal Routes

The signal routes are arbitrary and installed locally to the SRXs. However, do not use active prefixes in your environment as this may cause unwanted connectivity issues with those production prefixes. The signal routes used are 169.254.200.1/32 (active) and 169.254.200.2 (backup).

```

1  set chassis high-availability services-redundancy-group 1 active-signal-route 169.254.200.1
2  set chassis high-availability services-redundancy-group 1 backup-signal-route 169.254.200.2

```

(Signal route configuration)

The active (highest activeness priority) will install the “active” signal route while the backup (lowest activeness priority) will install the “backup” signal route in the respective local routing tables.

The active SRX (SRX-1) installs the active signal route 169.254.200.1 and does not have an entry for the backup signal route 169.254.200.2.

```

1  MNHA-SRX-1> show route 169.254.200.0/30
2
3  inet.0: 43 destinations, 70 routes (43 active, 0 holddown, 0 hidden)
4  + = Active Route, - = Last Active, * = Both
5
6  169.254.200.1/32 *[Static/12] 16:24:08
7                      Receive
8
9  MNHA-SRX-1>

```

(SRX-1 active signal route)

The backup SRX (SRX-2) installs the backup signal route 169.254.200.2 and does not have an entry for the active signal route 169.254.200.1.

```

1  MNHA-SRX-2> show route 169.254.200.0/30
2
3  inet.0: 41 destinations, 66 routes (41 active, 0 holddown, 0 hidden)
4  + = Active Route, - = Last Active, * = Both
5
6  169.254.200.2/32 *[Static/12] 16:24:47
7                      Receive
8
9  MNHA-SRX-2>

```

(SRX-2 backup signal route)

1.2.5 BGP Export Policy

The export policy will advertise the 192.168.252.0/24 prefix to respective BGP peers with specified MED if the active route (10) or backup route (20) are present in the routing table.

The policy options and conditions are configured identically on the SRXs.

```

1 MNHA-SRX-1> show configuration policy-options condition ACTIVE_ROUTE_EXISTS_SRG1 | display set
2 set policy-options condition ACTIVE_ROUTE_EXISTS_SRG1 if-route-exists address-family inet
   169.254.200.1/32
3 set policy-options condition ACTIVE_ROUTE_EXISTS_SRG1 if-route-exists address-family inet table
   inet.0
4
5 MNHA-SRX-1> show configuration policy-options condition BACKUP_ROUTE_EXISTS_SRG1 | display set
6 set policy-options condition BACKUP_ROUTE_EXISTS_SRG1 if-route-exists address-family inet
   169.254.200.2/32
7 set policy-options condition BACKUP_ROUTE_EXISTS_SRG1 if-route-exists address-family inet table
   inet.0

```

(Policy Options)

```

1 MNHA-SRX-1> show configuration | display set | grep MNHA_
2 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from protocol direct
3 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from route-filter 192.168.252.0/24
   exact
4 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from condition
   ACTIVE_ROUTE_EXISTS_SRG1
5 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 then metric 10
6 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 then accept
7 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from protocol direct
8 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from route-filter 192.168.252.0/24
   exact
9 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from condition
   BACKUP_ROUTE_EXISTS_SRG1
10 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 then metric 20
11 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 then accept
12 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 from protocol direct
13 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 then metric 30
14 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 then accept
15 set policy-options policy-statement MNHA_ROUTE_POLICY term default then reject
16 set protocols bgp group trust export MNHA_ROUTE_POLICY

```

(Export Policy)

RTR-1 preferred route to 192.168.252.0/24 is to SRX-1 (active – best metric) and RTR-2 preferred route to 192.168.252.0/24 is also to SRX-1 (active – best metric). If the link (BGP peers) from either RTR-1 or RTR-2 to SRX-1 fails, traffic will be routed across the cross connection between RTR-1 and RTR-2 to reach SRX-1.

```

1 MNHA-RTR-1> show route 192.168.252.0/24 receive-protocol bgp 10.10.99.5
2
3 inet.0: 45 destinations, 87 routes (45 active, 0 holddown, 0 hidden)
4   Prefix Nexthop MED Lclpref AS path
5   * 192.168.252.0/24 10.10.99.5 10 65022 I
6
7 MNHA-RTR-1> show route 192.168.252.0/24 receive-protocol bgp 10.10.99.131
8
9 inet.0: 45 destinations, 87 routes (45 active, 0 holddown, 0 hidden)
10  Prefix Nexthop MED Lclpref AS path
11  192.168.252.0/24 10.10.99.131 20 65022 I
12
13 MNHA-RTR-1> show route 192.168.252.0/24
14
15 inet.0: 45 destinations, 87 routes (45 active, 0 holddown, 0 hidden)
16 + = Active Route, - = Last Active, * = Both
17
18 192.168.252.0/24 *[BGP/170] 16:31:53, MED 10, localpref 100
19     AS path: 65022 I, validation-state: unverified
20     > to 10.10.99.5 via ge-0/0/0.0
21     [BGP/170] 00:17:46, MED 10, localpref 100
22     AS path: 65022 I, validation-state: unverified
23     > to 10.10.101.2 via ge-0/0/3.101
24     [BGP/170] 16:31:22, MED 20, localpref 100
25     AS path: 65022 I, validation-state: unverified
26     > to 10.10.99.131 via ge-0/0/2.0

```

(RTR-1 route to 192.168.252.0/24)

```

1 MNHA-RTR-2> show route 192.168.252.0/24 receive-protocol bgp 10.10.99.69
2
3 inet.0: 45 destinations, 87 routes (45 active, 0 holddown, 0 hidden)

```

```

4 Prefix Nexthop MED Lclpref AS path
5 * 192.168.252.0/24 10.10.99.69 10 65022 I
6
7 MNHA-RTR-2> show route 192.168.252.0/24 receive-protocol bgp 10.10.99.195
8
9 inet.0: 45 destinations, 87 routes (45 active, 0 holddown, 0 hidden)
10 Prefix Nexthop MED Lclpref AS path
11 192.168.252.0/24 10.10.99.195 20 65022 I
12
13 MNHA-RTR-2> show route 192.168.252.0/24
14
15 inet.0: 45 destinations, 87 routes (45 active, 0 holddown, 0 hidden)
16 + = Active Route, - = Last Active, * = Both
17
18 192.168.252.0/24 *[BGP/170] 00:19:32, MED 10, localpref 100
19 AS path: 65022 I, validation-state: unverified
20 > to 10.10.99.69 via ge-0/0/2.0
21 [BGP/170] 16:33:38, MED 10, localpref 100
22 AS path: 65022 I, validation-state: unverified
23 > to 10.10.101.1 via ge-0/0/3.101
24 [BGP/170] 15:04:14, MED 20, localpref 100
25 AS path: 65022 I, validation-state: unverified
26 > to 10.10.99.195 via ge-0/0/0.0

```

(RTR-2 route to 192.168.252.0/24)

1.3 MNHA – L2 Default Gateways

The active SRX will use 192.168.253.1 for VLAN 500 and will use 192.168.252.254 for the north bound L2 hand-off. A failover event will trigger a GARP advertisement to update the connected switching tables when the IP moves from the previously active SRX to the newly active SRX.

```

1 set chassis high-availability services-redundancy-group 1 virtual-ip 1 ip 192.168.252.254/24
2 set chassis high-availability services-redundancy-group 1 virtual-ip 1 interface ge-0/0/2.0
3 set chassis high-availability services-redundancy-group 1 virtual-ip 2 ip 192.168.253.1/24
4 set chassis high-availability services-redundancy-group 1 virtual-ip 2 interface ge-0/0/0.500

```

(VIP configuration)

The active SRX (SRX-1) will install the VIP while the backup SRX will show that the VIP is not installed.

```

1 Virtual IP Info:
2 Index: 2
3 IP: 192.168.253.1/24
4 VMAC: N/A
5 Interface: ge-0/0/0.500
6 Status: INSTALLED
7
8 Index: 1
9 IP: 192.168.252.254/24
10 VMAC: N/A
11 Interface: ge-0/0/2.0
12 Status: INSTALLED

```

(SRX-1 – Active)

```

1 Virtual IP Info:
2 Index: 2
3 IP: 192.168.253.1/24
4 VMAC: N/A
5 Interface: ge-0/0/0.500
6 Status: NOT INSTALLED
7
8 Index: 1
9 IP: 192.168.252.254/24
10 VMAC: N/A
11 Interface: ge-0/0/2.0
12 Status: NOT INSTALLED

```

(SRX-2 – Backup)

You can use a virtual-mac that will float with the active IP address, otherwise, the physical MAC of the interface will be used.

```

1 MNHA-SRX-1> show interfaces ge-0/0/0 | grep address:
2   Current address: 00:0c:29:3d:90:62, Hardware address: 00:0c:29:3d:90:62
3
4 MNHA-SRX-1> show interfaces ge-0/0/2 | grep address:
5   Current address: 00:0c:29:3d:90:76, Hardware address: 00:0c:29:3d:90:76
6
7 MNHA-SRX-2> show interfaces ge-0/0/0 | grep address:
8   Current address: 00:0c:29:ed:4b:42, Hardware address: 00:0c:29:ed:4b:42
9
10 MNHA-SRX-2> show interfaces ge-0/0/2 | grep address:
11   Current address: 00:0c:29:ed:4b:56, Hardware address: 00:0c:29:ed:4b:56

```

(SRX-1 and SRX-2 Hardware Addresses)

Hosts within the L2 domains respective gateways pointing to the VIP addresses on the SRXs shown in Figure 5.

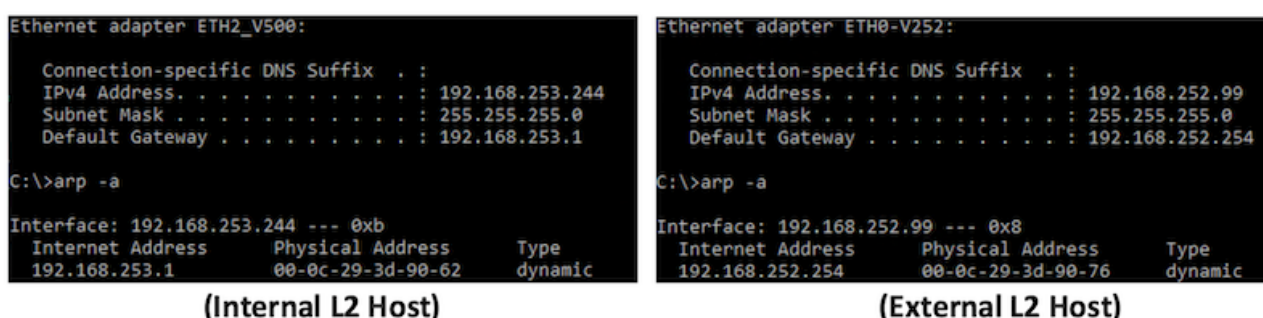


Figure 5: Hosts' Default Gateways

1.4 Monitoring and Failover Criteria

There are several combinations of monitoring thresholds and weights. For this example, BFD and interface monitoring is configured such that:

- If a single L3 BFD monitored interface fails, SRG failover will not occur (weight 50)
- If both L3 BFD monitored interfaces fails, SRG failover will occur (weight 100)
- If a single L2 interface fails, SRG failover will occur (weight 100)

```

1 [edit chassis high-availability services-redundancy-group 1]
2 MNHA-SRX-2# show | display set | grep monitor
3 ...monitor-object BFD_UPLINKS object-threshold 100
4 ...monitor-object BFD_UPLINKS bfd-liveliness threshold 100
5 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.193 src-ip 10.10.99.195
6 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.193 session-type singlehop
7 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.193 interface ge-0/0/0.10
8 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.193 weight 50
9 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.129 src-ip 10.10.99.131
10 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.129 session-type singlehop
11 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.129 interface ge-0/0/1.0
12 ...monitor-object BFD_UPLINKS bfd-liveliness destination-ip 10.10.99.129 weight 50
13 ...monitor-object L2-GWS object-threshold 100
14 ...monitor-object L2-GWS interface threshold 100
15 ...monitor-object L2-GWS interface interface-name ge-0/0/2 weight 100
16 ...monitor-object L2-GWS interface interface-name ge-0/0/1 weight 100
17 ...monitor-srg-threshold 100

```

(SRG Monitored Objects)

1.5 Traffic Flows

Traffic flows during a normalized state, with SRX-1 as the active node, traffic will first hit the hosts' respective default gateways. For the MNHA-L3 flow (Figure 6), the SRX-1 has an ECMP route to the destination and flows will be routed accordingly.

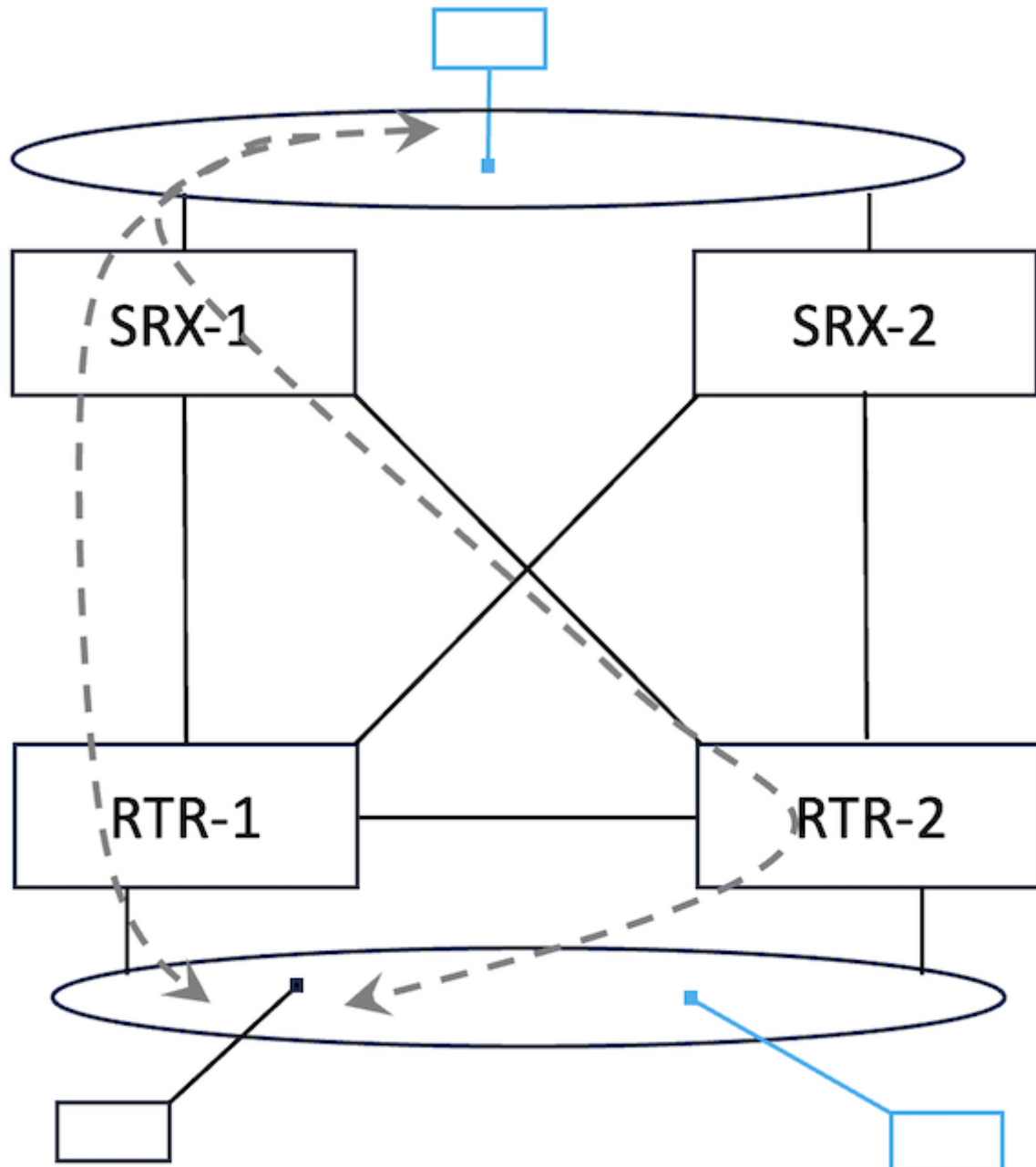


Figure 6: MNHA-L3 Traffic Flow

With the MNHA-DFG (Figure 7) scenario, return traffic (or originating traffic) from 192.168.253.244 will be sent to its default gateway on the active SRX.

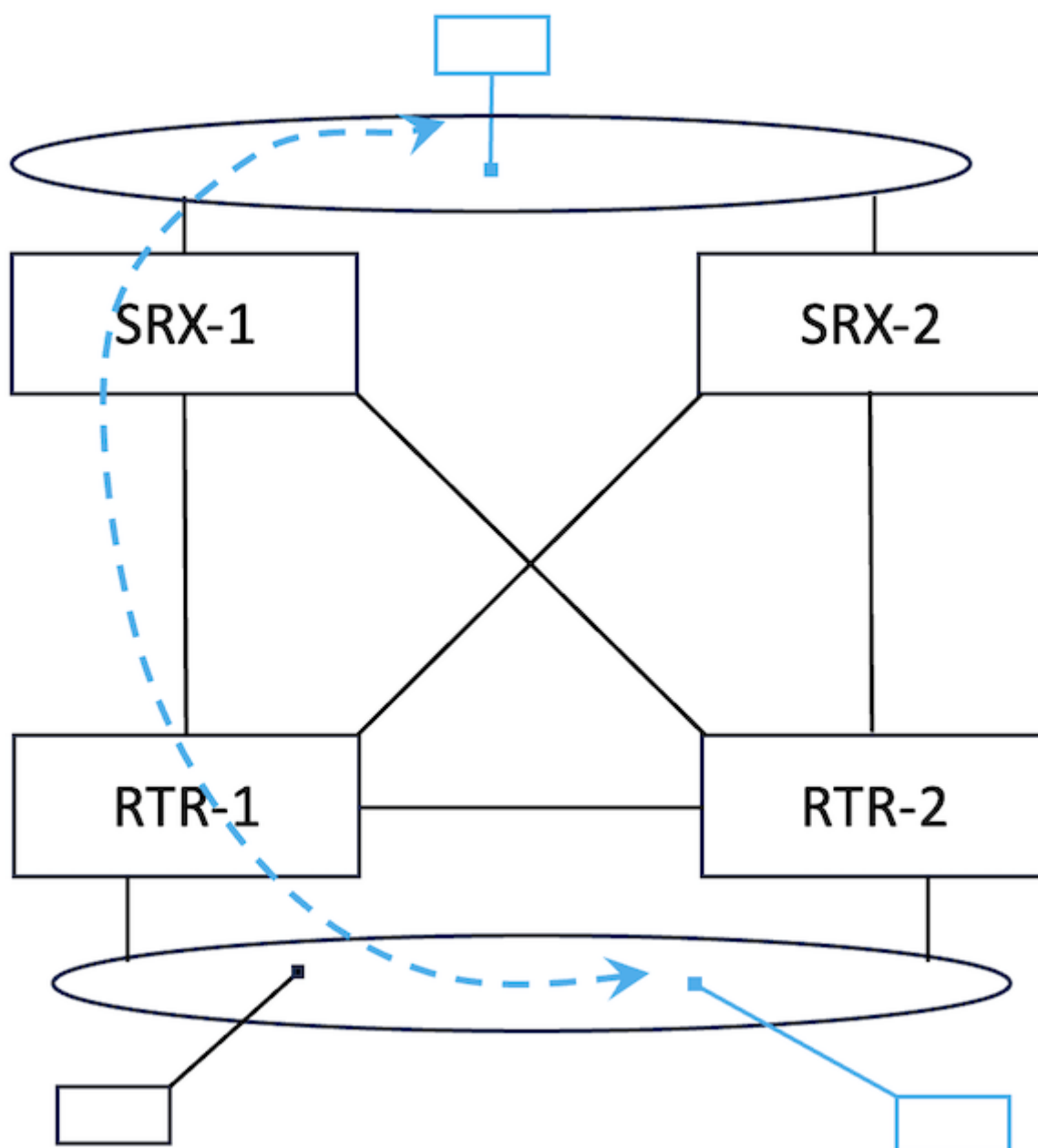


Figure 7: MNHA-DFG Traffic Flow

During failover testing, a **L3 flow** is represented as a long-lived (TCP 5202) session from 192.168.99.100, internal - behind RTR-X, to 192.168.252.99, external. An** L2 flow** is represented as a long-lived (TCP 5201) session from 192.168.253.244, internal, to 192.168.252.99, external.

Sessions on the active SRX will be installed in the table with an HA Wing state of Active. Session information, under normalized conditions:

```

1 MNHA-SRX-1> show security flow session destination-prefix 192.168.252.99/32 protocol tcp
2
3 Session ID: 118631, Policy name: default-policy-logical-system-00/2, HA State: Active, Timeout:
  1800, Session State: Valid
4 In: 192.168.253.244/55293 --> 192.168.252.99/5202;tcp, Conn Tag: 0x0, If: ge-0/0/0.500, Pkts:
  565563, Bytes: 846279017, HA Wing State: Active,
5 Out: 192.168.252.99/5202 --> 192.168.253.244/55293;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
  93439, Bytes: 3737596, HA Wing State: Active,
6
7 Session ID: 119130, Policy name: default-policy-logical-system-00/2, HA State: Active, Timeout:
  1800, Session State: Valid

```

```

8 In: 192.168.99.100/52650 --> 192.168.252.99/5201;tcp, Conn Tag: 0x0, If: ge-0/0/0.10, Pkts:
    244665, Bytes: 366100861, HA Wing State: Active,
9 Out: 192.168.252.99/5201 --> 192.168.99.100/52650;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
    23385, Bytes: 938184, HA Wing State: Active,
10 Total sessions: 2

```

(Active SRX Security Flows)

Sessions on the backup SRX will be installed in the table with an HA Wing state of Warm.

```

1 MNHA-SRX-2> show security flow session destination-prefix 192.168.252.99/32 protocol tcp
2 Session ID: 114931, Policy name: default-policy-logical-system-00/2, HA State: Warm, Timeout:
    13604, Session State: Valid
3 In: 192.168.253.244/55293 --> 192.168.252.99/5202;tcp, Conn Tag: 0x0, If: ge-0/0/0.500, Pkts:
    0, Bytes: 0, HA Wing State: Warm,
4 Out: 192.168.252.99/5202 --> 192.168.253.244/55293;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
    0, Bytes: 0, HA Wing State: Warm,
5
6 Session ID: 115414, Policy name: default-policy-logical-system-00/2, HA State: Warm, Timeout:
    13942, Session State: Valid
7 In: 192.168.99.100/52650 --> 192.168.252.99/5201;tcp, Conn Tag: 0x0, If: ge-0/0/0.10, Pkts: 0,
    Bytes: 0, HA Wing State: Warm,
8 Out: 192.168.252.99/5201 --> 192.168.99.100/52650;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts: 0,
    Bytes: 0, HA Wing State: Warm,
9 Total sessions: 2

```

(Backup SRX Security Flows)

1.5.1 Active Forwarding Path – BFD Detection

If there is failure on the active forwarding path in the bow-tie design, traffic will be instantly routed across the cross connect between RTR-1 and RTR-2 (MED 10) and a failover event will not occur as shown below with the change of interface on session (ID 119130) from 0/0/0.10 to 0/0/1.0. The L2 forwarding paths were not changed (0/0/0.500).

```

1 MNHA-SRX-1> show bfd session
2
3 Address State Interface Time Interval Multiplier
4 10.10.99.1 Down ge-0/0/0.10 1.500 2.000 3
5 10.10.99.65 Up ge-0/0/1.0 1.500 0.500 3
6 100.0.0.1 Up 2.000 0.400 5
7
8 3 sessions, 5 clients
9 Cumulative transmit rate 5.0 pps, cumulative receive rate 6.5 pps
10
11 jadmin@MNHA-SRX-1> show security flow session destination-prefix 192.168.252.99/32 protocol tcp
12 Session ID: 118631, Policy name: default-policy-logical-system-00/2, HA State: Active, Timeout:
    1800, Session State: Valid
13 In: 192.168.253.244/55293 --> 192.168.252.99/5202;tcp, Conn Tag: 0x0, If: ge-0/0/0.500, Pkts:
    1166223, Bytes: 1745079945, HA Wing State: Active,
14 Out: 192.168.252.99/5202 --> 192.168.253.244/55293;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
    186388, Bytes: 7455580, HA Wing State: Active,
15
16 Session ID: 119130, Policy name: default-policy-logical-system-00/2, HA State: Active, Timeout:
    1800, Session State: Valid
17 In: 192.168.99.100/52650 --> 192.168.252.99/5201;tcp, Conn Tag: 0x0, If: ge-0/0/1.0, Pkts:
    854061, Bytes: 1277977581, HA Wing State: Active,
18 Out: 192.168.252.99/5201 --> 192.168.99.100/52650;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
    79932, Bytes: 3212636, HA Wing State: Active,
19 Total sessions: 2

```

(Active forwarding path failure, SRX-1)

With an active fault condition, single BFD session down between SRX-1 and RTR-1, the backup SRX session table remains unchanged; no failover event triggered.

```

1 MNHA-SRX-2> show security flow session destination-prefix 192.168.252.99/32 protocol tcp
2 Session ID: 114931, Policy name: default-policy-logical-system-00/2, HA State: Warm, Timeout:
    13070, Session State: Valid

```

```

3 In: 192.168.253.244/55293 --> 192.168.252.99/5202;tcp, Conn Tag: 0x0, If: ge-0/0/0.500, Pkts:
  0, Bytes: 0, HA Wing State: Warm,
4 Out: 192.168.252.99/5202 --> 192.168.253.244/55293;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
  0, Bytes: 0, HA Wing State: Warm,
5
6 Session ID: 115414, Policy name: default-policy-logical-system-00/2, HA State: Warm, Timeout:
  13408, Session State: Valid
7 In: 192.168.99.100/52650 --> 192.168.252.99/5201;tcp, Conn Tag: 0x0, If: ge-0/0/0.10, Pkts: 0,
  Bytes: 0, HA Wing State: Warm,
8 Out: 192.168.252.99/5201 --> 192.168.99.100/52650;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts: 0,
  Bytes: 0, HA Wing State: Warm,

```

(Active forwarding path failure, SRX-2)

1.5.2 Failover Event – BFD Detection

If both redundant BFD sessions fail, a failover event is triggered (Figure 8). The initially active SRX (SRX-1) transitions into an Ineligible state while the previously backup SRX (SRX-2) moves into the active state. The sessions transition from Warm to Active on the newly active SRX-2 and vice versa for SRX-1.

```

1 MNHA-SRX-1> show bfd session
2                                     Detect Transmit
3 Address State Interface Time Interval Multiplier
4 10.10.99.1 Down ge-0/0/0.10 1.500 2.000 3
5 10.10.99.65 Down ge-0/0/1.0 1.500 2.000 3
6 100.0.0.1 Up 2.000 0.400 5
7
8 3 sessions, 5 clients
9 Cumulative transmit rate 3.5 pps, cumulative receive rate 6.5 pps
10
11 jadmin@MNHA-SRX-1> show security flow session destination-prefix 192.168.252.99/32 protocol tcp
12 Session ID: 118631, Policy name: default-policy-logical-system-00/2, HA State: Warm, Timeout:
  1808, Session State: Valid
13 In: 192.168.253.244/55293 --> 192.168.252.99/5202;tcp, Conn Tag: 0x0, If: ge-0/0/0.500, Pkts:
  1507953, Bytes: 2256429529, HA Wing State: Warm,
14 Out: 192.168.252.99/5202 --> 192.168.253.244/55293;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
  240892, Bytes: 9635740, HA Wing State: Warm,
15
16 Session ID: 119130, Policy name: default-policy-logical-system-00/2, HA State: Warm, Timeout:
  1806, Session State: Valid
17 In: 192.168.99.100/52650 --> 192.168.252.99/5201;tcp, Conn Tag: 0x0, If: ge-0/0/1.0, Pkts:
  1202249, Bytes: 1798992517, HA Wing State: Warm,
18 Out: 192.168.252.99/5201 --> 192.168.99.100/52650;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
  111920, Bytes: 4496748, HA Wing State: Warm,
19 Total sessions: 2

```

(BFD Triggered Failover – SRX-1)

```

1 MNHA-SRX-2> show bfd session
2                                     Detect Transmit
3 Address State Interface Time Interval Multiplier
4 10.10.99.129 Up ge-0/0/1.0 1.500 0.500 3
5 10.10.99.193 Up ge-0/0/0.10 1.500 0.500 3
6 100.0.0.0 Up 2.000 0.400 5
7
8 3 sessions, 5 clients
9 Cumulative transmit rate 6.5 pps, cumulative receive rate 6.5 pps
10
11 jadmin@MNHA-SRX-2> show security flow session destination-prefix 192.168.252.99/32 protocol tcp
12 Session ID: 114931, Policy name: default-policy-logical-system-00/2, HA State: Active, Timeout:
  1798, Session State: Valid
13 In: 192.168.253.244/55293 --> 192.168.252.99/5202;tcp, Conn Tag: 0x0, If: ge-0/0/0.500, Pkts:
  26550, Bytes: 39728240, HA Wing State: Active,
14 Out: 192.168.252.99/5202 --> 192.168.253.244/55293;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
  4299, Bytes: 171960, HA Wing State: Active,
15
16 Session ID: 115414, Policy name: default-policy-logical-system-00/2, HA State: Active, Timeout:
  1798, Session State: Valid
17 In: 192.168.99.100/52650 --> 192.168.252.99/5201;tcp, Conn Tag: 0x0, If: ge-0/0/0.10, Pkts:
  26291, Bytes: 39337804, HA Wing State: Active,
18 Out: 192.168.252.99/5201 --> 192.168.99.100/52650;tcp, Conn Tag: 0x0, If: ge-0/0/2.0, Pkts:
  2998, Bytes: 123396, HA Wing State: Active,
19 Total sessions: 2

```

(BFD Triggered Failover – SRX-2)

Figure 8 visualizes the BFD detection time, impacting L3 traffic for 1.5 seconds.

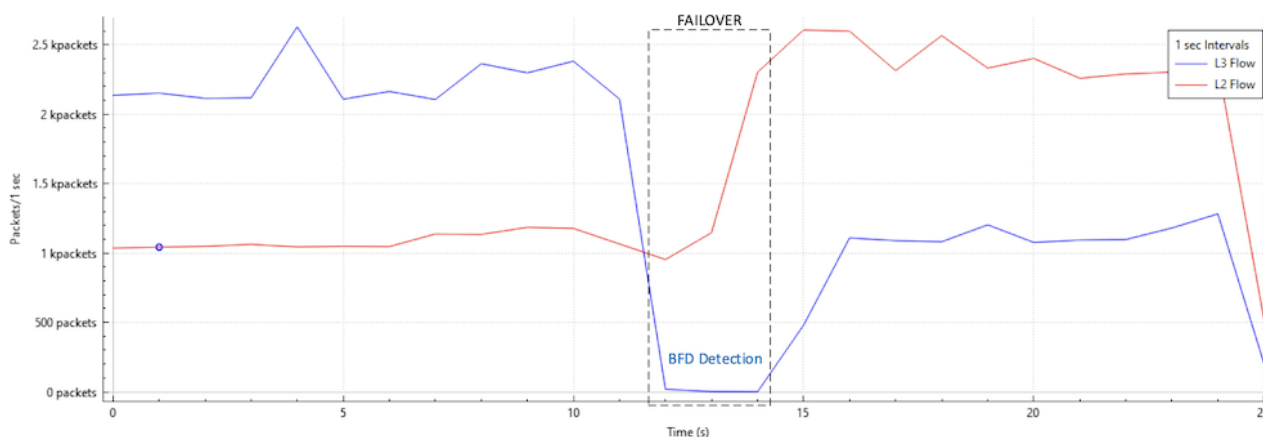


Figure 8: BFD triggered Failover

1.5.3 Failover Event – Reboot Active Node and Split-Brain Detection

With hybrid deployments, when there is impact to the ICL connectivity, for example losing power to the active node, the mechanism to tell the backup SRX for an SRG group to transition to active is gone. Once the configurable detection time expires, 2 seconds in these scenarios, either an ICMP or BFD based split-brain detection initializes. Split-brain detection is leveraged to ensure that there is not an opportunity for the VIP to be active on both SRXs (duplicate IP address) and negatively impacting traffic flows.

ICMP activeness is the default method for split-brain detection. The following example uses a 1 second probe, requiring 3 failed responses to declare the SRX as active for the VIP – triggering the GARP process (if previously backup for the SRG). The default gateway, VIP, move can have an impact to traffic between 3-8 seconds with ICMP detection.

```
1 set chassis high-availability services-redundancy-group 1 activeness-probe minimum-interval 1000
2 set chassis high-availability services-redundancy-group 1 activeness-probe multiplier 3
```

(ICMP Activeness Configuration)



Figure 9: Reboot Active SRG SRX ICMP Based

Notice that L3 traffic, traffic routed towards the destination versus switched, was not impacted. The interfaces

were hard down during the reboot, traffic routed from SRX-1 (00:0C:29:3D:90:76) to 192.168.252.99 via SRX-2 (00:0C:29:ED:4B:56) (Figure 10).

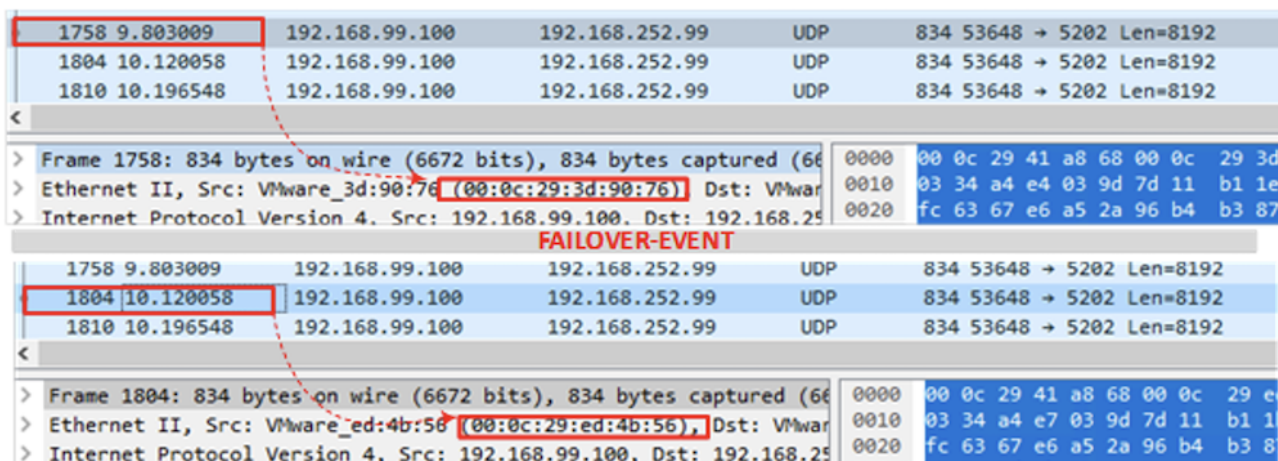


Figure 10: Reboot Active SRG SRX - L3 flow

Single-hop BFD is an alternative to ICMP based split-brain detection. While ICMP based split-brain detection initiates after an ICL failure is detected, BFD is constantly running on the SRG and can achieve quicker default gateway failovers. Be aware that you can not configure both ICMP and BFD split-brain detections simultaneously.

Additional details about [Split-Brain Detection](#).

1.5.4 Preemption

With preemption, there is an automatic fail-back once the monitoring criteria is met. The BFD BGP hold-down timer mentioned previously is simply used to illustrate potential convergence/instability issues along the active SRXs forwarding path(s). In the following example (Figure 11), SRX-1 preempts becoming active. L2 traffic is not impacted (with interface failover). However, the L3 traffic is impacted by the duration of the hold-down timer (10s); the BGP peers are not up while SRX-1 is active.

Failover event graphs were from packet captures taken on 192.168.252.99 (L2 external host).

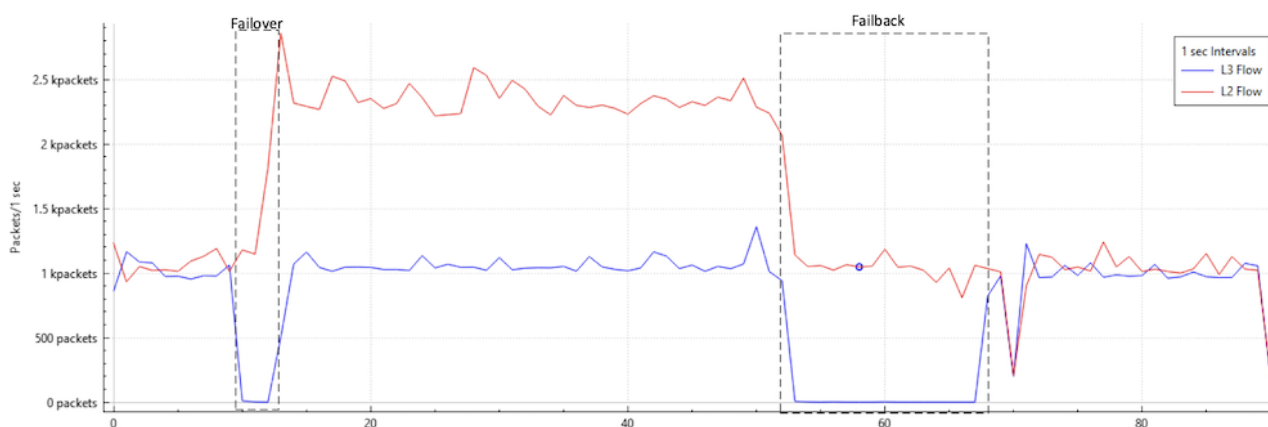


Figure 11: Preempt Failover/Failback with hold-down

Figure 12 shows the same traffic flows during failover and a preempted failback, without the BGP BFD hold-down timer configured.

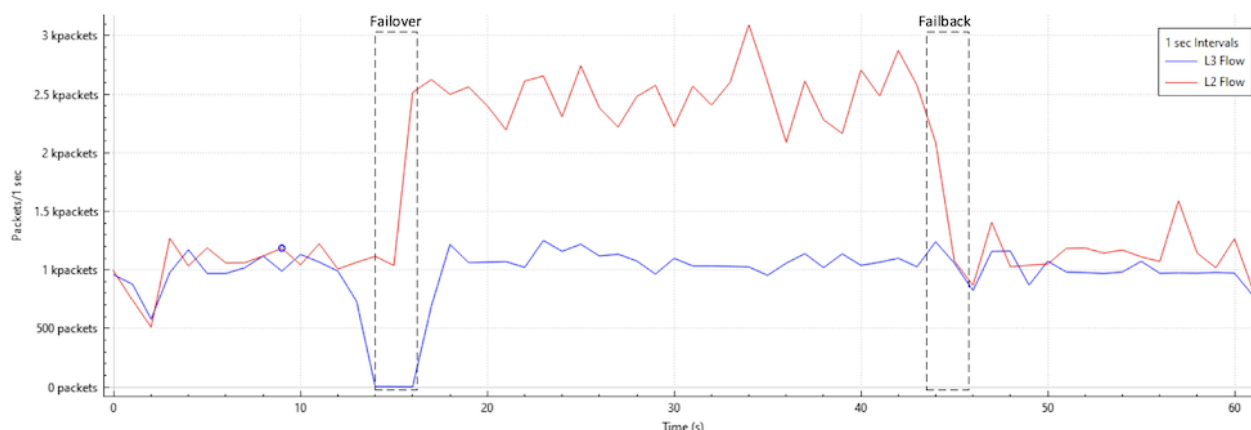


Figure 12: Preempt Failover/Failback without hold-down

```
1 request chassis high-availability failover peer-id 1 services-redundancy-group 1
```

Figure 13 shows the same failover, with BGP BFD hold down (10s) configured, without preemption and a manual fail back to SRX-1 from SRX-2.

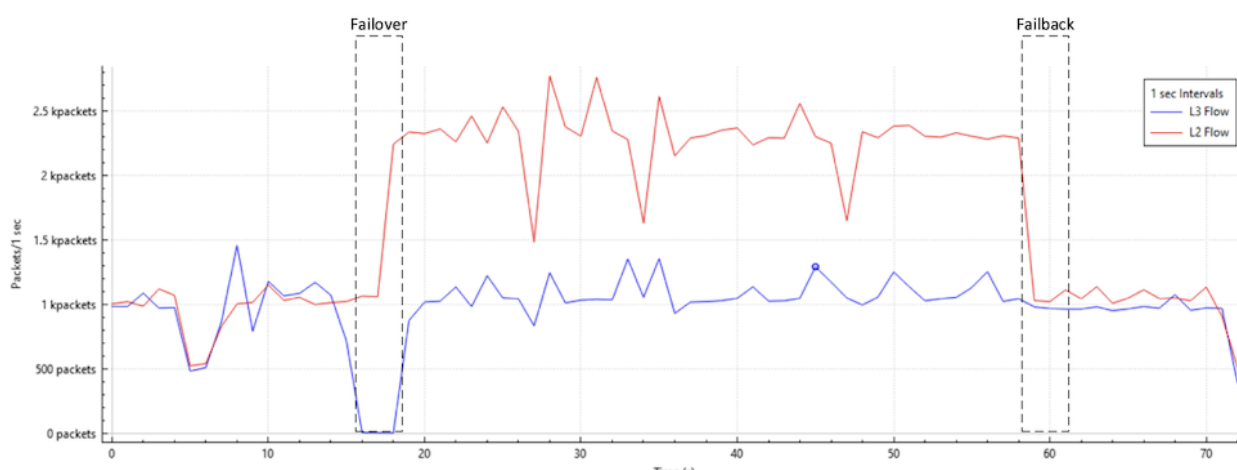


Figure 13: Failover/Manual Failback with hold-down

1.6 Alternative Designs

1.6.1 iBGP Considerations

Under normalized operations, no failed links, a bow-tie design with iBGP and MNHA operates as expected – symmetrical traffic to the active SRX. However, there may be an issue during an active failed link event without any additional design considerations. For example, if there was a BGP peer failure on the active forwarding path, let's say RTR-1 to SRX-1, traffic will not flow across the inter-connect between RTR-1 and RTR-2 to reach SRX-1. Instead, traffic will flow from the link between RTR-1 to SRX-2 (backup SRX) creating an asymmetric flow and causing stateful flows to break. This is due to BGP's loop prevention, preventing iBGP peers from forwarding iBGP learned prefixes (SRX-1/2) to other iBGP neighbors (RTR-1/2).

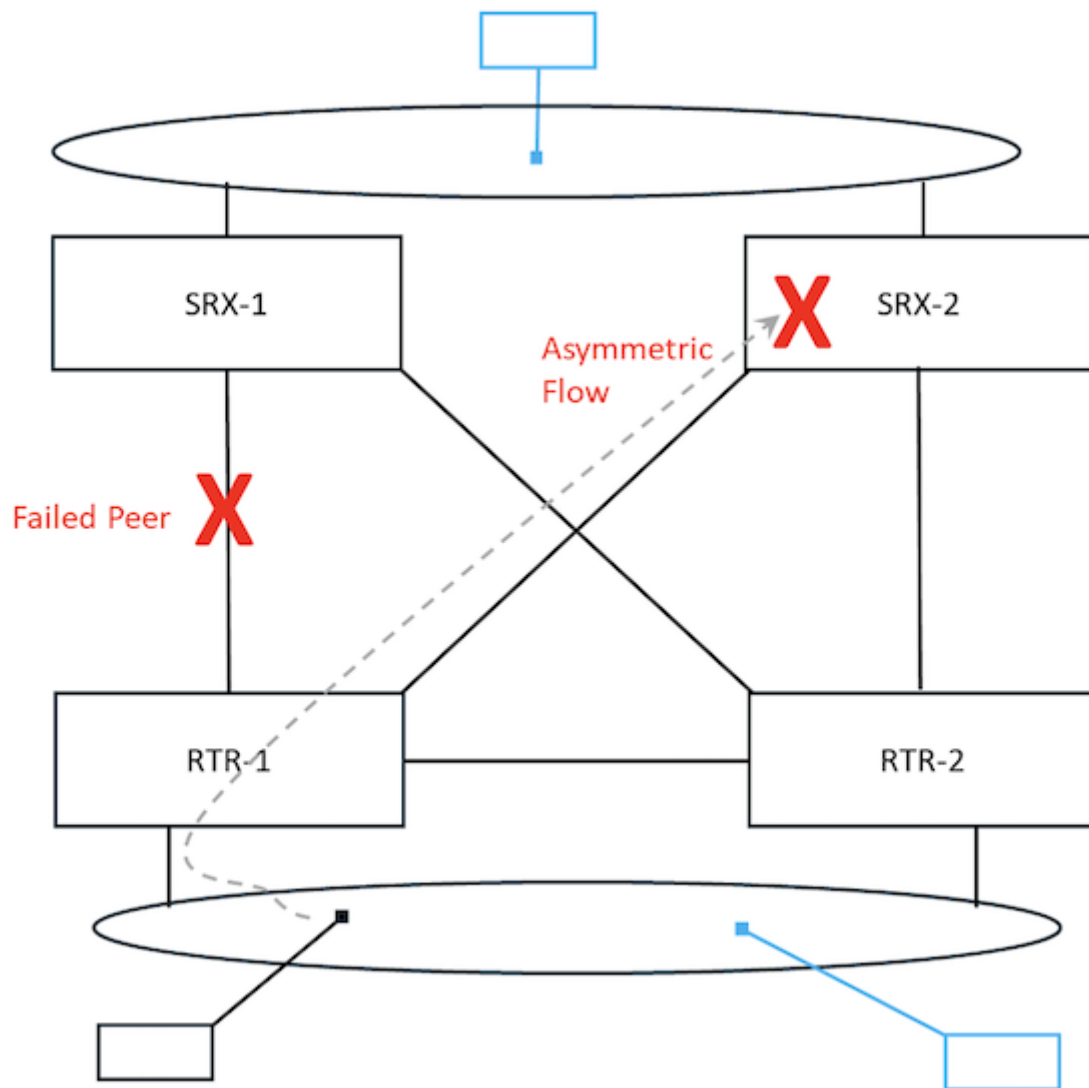


Figure 14: Asymmetrical Traffic Flow

To mitigate stateful traffic disruptions during an active failed path, with an iBGP bowtie configuration, implement an IGP with appropriate redistribution or a “floating static route” between RTR-1 and RTR-2. Another mitigation method is to use [ICD \(Inter Chassis Datapath\)](#) link(s).

1.6.2 LAG/No Bow-tie

The same level of physical redundancy can be achieved by deploying 802.11ad (LACP) aggregate interfaces (Figure 15) between the routers and SRXs without “crossing” the links. Instead of ECMP load-balancing, the load-balancing would be with the LAG’s hashing algorithm.

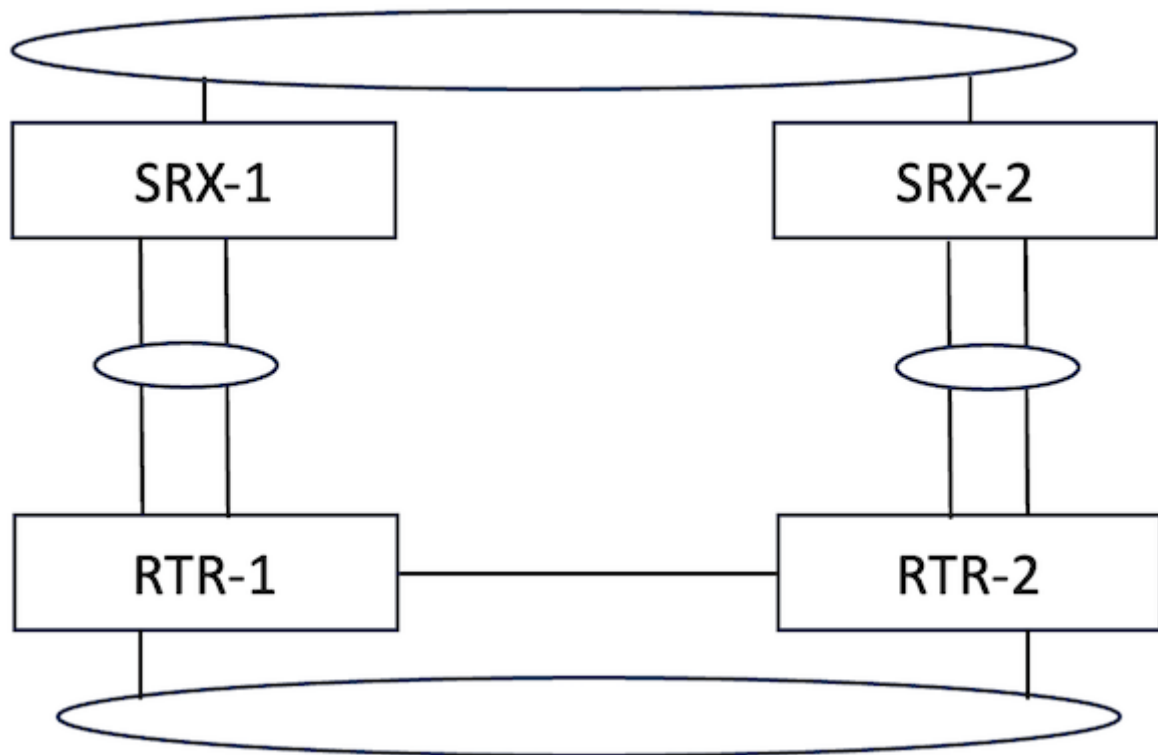


Figure 15: LAG Uplinks

1.6.3 Active/Active Service Redundancy Groups

Another alternative would be to use 2 SRGs (Service Redundancy Groups) in an active/active (Figure 16) scenario such that the redundant L3 links from the routers are split between the 2 groups. This would also require an additional VIP.

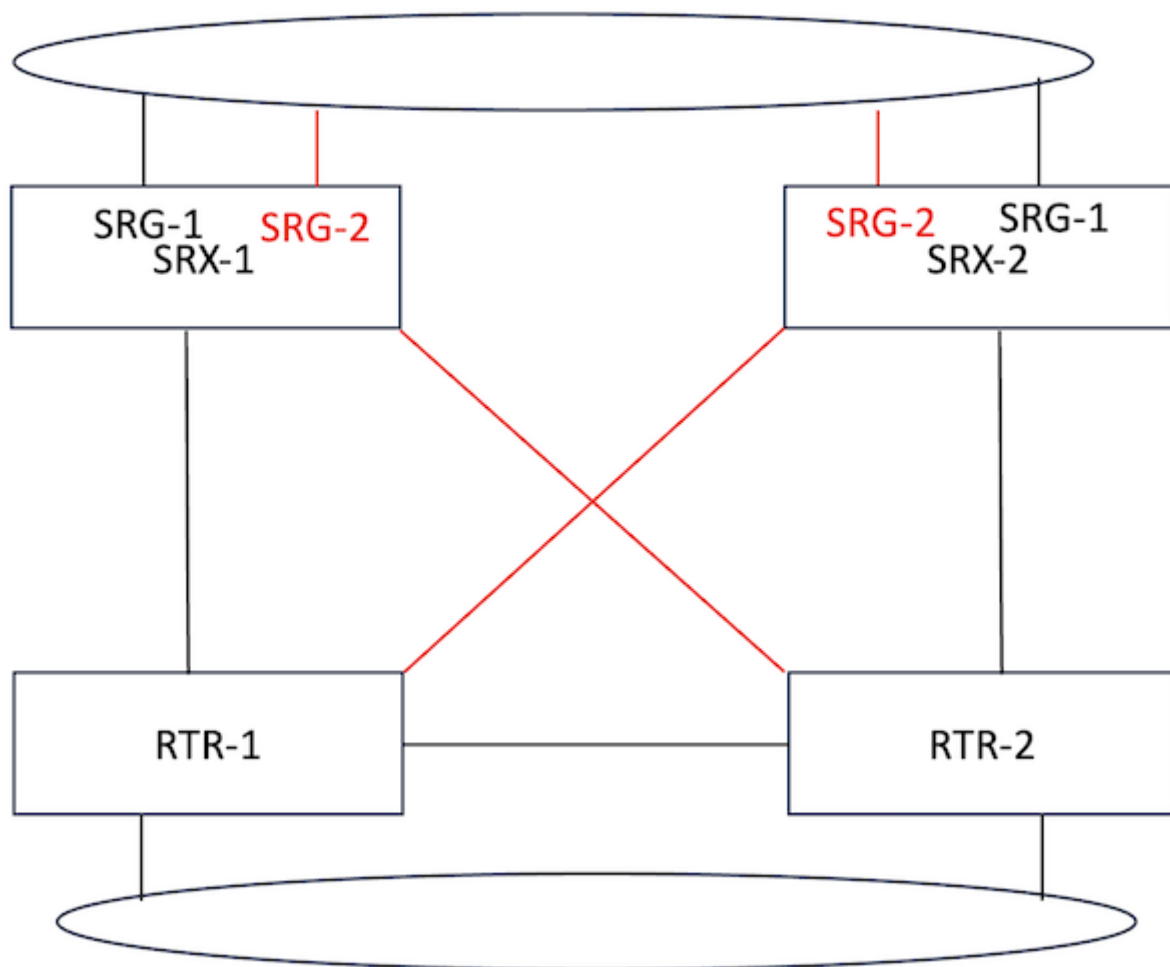


Figure 16: Active-Active Multiple SRG

1.7 Configurations

1.7.1 RTR-1

```

1 set system host-name MNHA-RTR-1
2 set system no-redirects
3 set security forwarding-options family inet mode packet-based
4 set interfaces ge-0/0/0 unit 0 family inet address 10.10.99.1/26
5 set interfaces ge-0/0/1 unit 0 family inet address 192.168.99.178/24
6 set interfaces ge-0/0/2 unit 0 family inet address 10.10.99.129/26
7 set interfaces ge-0/0/3 flexible-vlan-tagging
8 set interfaces ge-0/0/3 unit 101 vlan-id 101
9 set interfaces ge-0/0/3 unit 101 family inet address 10.10.101.1/24
10 set interfaces ge-0/0/3 unit 201 vlan-id 201
11 set interfaces ge-0/0/3 unit 201 family inet address 10.10.201.1/24
12 set interfaces ge-0/0/4 flexible-vlan-tagging
13 set interfaces ge-0/0/4 native-vlan-id 500
14 set interfaces ge-0/0/4 unit 0 family ethernet-switching interface-mode trunk set interfaces ge
-0/0/4 unit 0 family ethernet-switching vlan members V500
15 set interfaces ge-0/0/5 flexible-vlan-tagging
16 set interfaces ge-0/0/5 unit 0 family ethernet-switching interface-mode trunk
17 set interfaces ge-0/0/5 unit 0 family ethernet-switching vlan members V500
18 set interfaces fxp0 unit 0 family inet address 192.168.100.178/24
19 set interfaces lo0 unit 101 family inet address 10.11.11.12/32
20 set interfaces lo0 unit 201 family inet address 10.11.201.1/32
21 set policy-options policy-statement NHS term 1 from protocol bgp
22 set policy-options policy-statement NHS term 1 from route-type external
23 set policy-options policy-statement NHS term 1 then next-hop self
24 set policy-options policy-statement load-balancing-policy then load-balance per-flow
25 set protocols bgp group EXT type external
26 set protocols bgp group EXT multihop ttl 4

```

```

27 set protocols bgp group EXT local-address 192.168.99.178
28 set protocols bgp group EXT local-as 65012
29 set protocols bgp group EXT multipath multiple-as
30 set protocols bgp group EXT bfd-liveness-detection minimum-interval 500
31 set protocols bgp group EXT bfd-liveness-detection multiplier 3
32 set protocols bgp group EXT neighbor 192.168.99.1 peer-as 65002
33 set protocols bgp group INT type internal
34 set protocols bgp group INT export NHS
35 set protocols bgp group INT local-as 65012
36 set protocols bgp group INT bfd-liveness-detection minimum-interval 500
37 set protocols bgp group INT bfd-liveness-detection minimum-receive-interval 500
38 set protocols bgp group INT bfd-liveness-detection multiplier 3
39 set protocols bgp group INT neighbor 10.10.101.2
40 set protocols bgp group INT-ebgp type external
41 set protocols bgp group INT-ebgp local-as 65012
42 set protocols bgp group INT-ebgp bfd-liveness-detection minimum-interval 500
43 set protocols bgp group INT-ebgp bfd-liveness-detection multiplier 3
44 set protocols bgp group INT-ebgp bfd-liveness-detection holddown-interval 10000
45 set protocols bgp group INT-ebgp neighbor 10.10.99.131 peer-as 65022
46 set protocols bgp group INT-ebgp neighbor 10.10.99.5 peer-as 65022
47 set routing-options static route 192.168.99.100/32 next-hop 192.168.99.1
48 set routing-options forwarding-table export load-balancing-policy
49 set vlans V500 vlan-id 500
50 set vlans VLAN101 vlan-id 101

```

1.7.2 RTR-2

```

1 set system host-name MNHA-RTR-2
2 set system no-redirects
3 set security forwarding-options family inet mode packet-based
4 set interfaces ge-0/0/0 unit 0 family inet address 10.10.99.193/26
5 set interfaces ge-0/0/1 unit 0 family inet address 192.168.99.179/24
6 set interfaces ge-0/0/2 unit 0 family inet address 10.10.99.65/26
7 set interfaces ge-0/0/3 flexible-vlan-tagging
8 set interfaces ge-0/0/3 unit 101 vlan-id 101
9 set interfaces ge-0/0/3 unit 101 family inet address 10.10.101.2/24
10 set interfaces ge-0/0/3 unit 201 vlan-id 201
11 set interfaces ge-0/0/3 unit 201 family inet address 10.10.201.2/24
12 set interfaces ge-0/0/4 flexible-vlan-tagging
13 set interfaces ge-0/0/4 native-vlan-id 500
14 set interfaces ge-0/0/4 unit 0 family ethernet-switching interface-mode trunk
15 set interfaces ge-0/0/4 unit 0 family ethernet-switching vlan members V500
16 set interfaces ge-0/0/5 flexible-vlan-tagging
17 set interfaces ge-0/0/5 unit 0 family ethernet-switching interface-mode trunk
18 set interfaces ge-0/0/5 unit 0 family ethernet-switching vlan members V500
19 set interfaces fxp0 unit 0 family inet address 192.168.100.179/24
20 set interfaces lo0 unit 101 family inet address 10.11.11.11/32
21 set policy-options policy-statement NHS term 1 from protocol bgp
22 set policy-options policy-statement NHS term 1 from route-type external
23 set policy-options policy-statement NHS term 1 then next-hop self
24 set policy-options policy-statement load-balancing-policy then load-balance per-flow
25 set protocols bgp group EXT type external
26 set protocols bgp group EXT local-address 192.168.99.179
27 set protocols bgp group EXT local-as 65012
28 set protocols bgp group EXT multipath multiple-as
29 set protocols bgp group EXT bfd-liveness-detection minimum-interval 500
30 set protocols bgp group EXT bfd-liveness-detection minimum-receive-interval 500
31 set protocols bgp group EXT bfd-liveness-detection multiplier 3
32 set protocols bgp group EXT neighbor 192.168.99.1 peer-as 65002
33 set protocols bgp group INT type internal
34 set protocols bgp group INT export NHS
35 set protocols bgp group INT local-as 65012
36 set protocols bgp group INT bfd-liveness-detection minimum-interval 500
37 set protocols bgp group INT bfd-liveness-detection multiplier 3
38 set protocols bgp group INT neighbor 10.10.101.1
39 set protocols bgp group INT-ebgp local-as 65012
40 set protocols bgp group INT-ebgp bfd-liveness-detection minimum-interval 500
41 set protocols bgp group INT-ebgp bfd-liveness-detection multiplier 3
42 set protocols bgp group INT-ebgp bfd-liveness-detection holddown-interval 10000
43 set protocols bgp group INT-ebgp neighbor 10.10.99.69 peer-as 65022
44 set protocols bgp group INT-ebgp neighbor 10.10.99.195 peer-as 65022
45 set routing-options static route 192.168.99.100/32 next-hop 192.168.99.1
46 set routing-options forwarding-table export load-balancing-policy
47 set vlans V500 vlan-id 500
48 set vlans VLAN101 vlan-id 101

```

1.7.3 SRX-1

```

1 set system host-name MNHA-SRX-1
2 set chassis high-availability local-id 1
3 set chassis high-availability services-redundancy-group 1 activeness-probe minimum-interval 1000
4 set chassis high-availability services-redundancy-group 1 activeness-probe multiplier 3
5 set chassis high-availability local-id local-ip 100.0.0.0
6 set chassis high-availability peer-id 2 peer-ip 100.0.0.1
7 set chassis high-availability peer-id 2 interface ge-0/0/3.0
8 set chassis high-availability peer-id 2 liveness-detection minimum-interval 400
9 set chassis high-availability peer-id 2 liveness-detection multiplier 5
10 set chassis high-availability services-redundancy-group 0 peer-id 2
11 set chassis high-availability services-redundancy-group 1 deployment-type hybrid
12 set chassis high-availability services-redundancy-group 1 peer-id 2
13 set chassis high-availability services-redundancy-group 1 virtual-ip 1 ip 192.168.252.254/24
14 set chassis high-availability services-redundancy-group 1 virtual-ip 1 interface ge-0/0/2.0
15 set chassis high-availability services-redundancy-group 1 virtual-ip 2 ip 192.168.253.1/24
16 set chassis high-availability services-redundancy-group 1 virtual-ip 2 interface ge-0/0/0.500
17 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS
    object-threshold 100
18 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness threshold 100
19 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.1 src-ip 10.10.99.5
20 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.1 session-type singlehop
21 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.1 interface ge-0/0/0.10
22 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.1 weight 50
23 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.65 src-ip 10.10.99.69
24 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.65 session-type singlehop
25 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.65 interface ge-0/0/1.0
26 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.65 weight 50
27 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS object-
    threshold 100
28 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS
    interface threshold 100
29 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS
    interface interface-name ge-0/0/2 weight 100
30 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS
    interface interface-name ge-0/0/0 weight 100
31 set chassis high-availability services-redundancy-group 1 monitor srg-threshold 100
32 set chassis high-availability services-redundancy-group 1 active-signal-route 169.254.200.1
33 set chassis high-availability services-redundancy-group 1 backup-signal-route 169.254.200.2
34 set chassis high-availability services-redundancy-group 1 activeness-priority 200
35 set security policies default-policy permit-all
36 set security zones security-zone HA-ICL host-inbound-traffic system-services all
37 set security zones security-zone HA-ICL host-inbound-traffic protocols all
38 set security zones security-zone HA-ICL interfaces ge-0/0/3.0
39 set security zones security-zone trust host-inbound-traffic system-services all
40 set security zones security-zone trust host-inbound-traffic protocols all
41 set security zones security-zone trust interfaces ge-0/0/1.0
42 set security zones security-zone trust interfaces ge-0/0/0.500
43 set security zones security-zone trust interfaces ge-0/0/0.10
44 set security zones security-zone untrust host-inbound-traffic system-services ping
45 set security zones security-zone untrust interfaces ge-0/0/2.0
46 set interfaces ge-0/0/0 flexible-vlan-tagging
47 set interfaces ge-0/0/0 native-vlan-id 10
48 set interfaces ge-0/0/0 unit 10 vlan-id 10
49 set interfaces ge-0/0/0 unit 10 family inet address 10.10.99.5/26
50 set interfaces ge-0/0/0 unit 500 vlan-id 500
51 set interfaces ge-0/0/0 unit 500 family inet address 192.168.253.2/24
52 set interfaces ge-0/0/1 unit 0 family inet address 10.10.99.69/26
53 set interfaces ge-0/0/2 unit 0 family inet address 192.168.252.252/24
54 set interfaces ge-0/0/3 mtu 1514
55 set interfaces ge-0/0/3 unit 0 family inet address 100.0.0.0/31
56 set interfaces fxp0 unit 0 family inet address 192.168.100.180/24
57 set interfaces lo0 unit 0 family inet address 172.16.0.1/32
58 set forwarding-options hash-key family inet layer-3
59 set forwarding-options hash-key family inet layer-4
60 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from protocol direct
61 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from route-filter 192.168.252.0/24
    exact

```

```

62 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from condition
    ACTIVE_ROUTE_EXISTS_SRG1
63 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 then metric 10
64 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 then accept
65 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from protocol direct
66 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from route-filter 192.168.252.0/24
    exact
67 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from condition
    BACKUP_ROUTE_EXISTS_SRG1
68 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 then metric 20
69 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 then accept
70 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 from protocol direct
71 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 then metric 30
72 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 then accept
73 set policy-options policy-statement MNHA_ROUTE_POLICY term default then reject
74 set policy-options policy-statement load-balancing-policy then load-balance per-flow
75 set policy-options condition ACTIVE_ROUTE_EXISTS_SRG1 if-route-exists address-family inet
    169.254.200.1/32
76 set policy-options condition ACTIVE_ROUTE_EXISTS_SRG1 if-route-exists address-family inet table
    inet.0
77 set policy-options condition BACKUP_ROUTE_EXISTS_SRG1 if-route-exists address-family inet
    169.254.200.2/32
78 set policy-options condition BACKUP_ROUTE_EXISTS_SRG1 if-route-exists address-family inet table
    inet.0
79 set protocols bgp group trust type external
80 set protocols bgp group trust export MNHA_ROUTE_POLICY
81 set protocols bgp group trust local-as 65022
82 set protocols bgp group trust multipath
83 set protocols bgp group trust bfd-liveness-detection minimum-interval 500
84 set protocols bgp group trust bfd-liveness-detection minimum-receive-interval 500
85 set protocols bgp group trust bfd-liveness-detection multiplier 3
86 set protocols bgp group trust bfd-liveness-detection holddown-interval 10000
87 set protocols bgp group trust neighbor 10.10.99.1 peer-as 65012
88 set protocols bgp group trust neighbor 10.10.99.65 peer-as 65012
89 set routing-options forwarding-table export load-balancing-policy

```

SRX-2

```

1 set system host-name MNHA-SRX-2
2 set chassis high-availability local-id 2
3 set chassis high-availability services-redundancy-group 1 activeness-probe minimum-interval 1000
4 set chassis high-availability services-redundancy-group 1 activeness-probe multiplier 3
5 set chassis high-availability local-id local-ip 100.0.0.1
6 set chassis high-availability peer-id 1 peer-ip 100.0.0.0
7 set chassis high-availability peer-id 1 interface ge-0/0/3.0
8 set chassis high-availability peer-id 1 liveness-detection minimum-interval 400
9 set chassis high-availability peer-id 1 liveness-detection multiplier 5
10 set chassis high-availability services-redundancy-group 0 peer-id 1
11 set chassis high-availability services-redundancy-group 1 deployment-type hybrid
12 set chassis high-availability services-redundancy-group 1 peer-id 1
13 set chassis high-availability services-redundancy-group 1 virtual-ip 1 ip 192.168.252.254/24
14 set chassis high-availability services-redundancy-group 1 virtual-ip 1 interface ge-0/0/2.0
15 set chassis high-availability services-redundancy-group 1 virtual-ip 2 ip 192.168.253.1/24
16 set chassis high-availability services-redundancy-group 1 virtual-ip 2 interface ge-0/0/0.500
17 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS
    object-threshold 100
18 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness threshold 100
19 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.193 src-ip 10.10.99.195
20 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.193 session-type singlehop
21 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.193 interface ge-0/0/0.10
22 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.193 weight 50
23 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.129 src-ip 10.10.99.131
24 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.129 session-type singlehop
25 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.129 interface ge-0/0/1.0
26 set chassis high-availability services-redundancy-group 1 monitor monitor-object BFD_UPLINKS bfd
    -liveness destination-ip 10.10.99.129 weight 50
27 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS object-
    threshold 100
28 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS
    interface threshold 100
29 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS
    interface interface-name ge-0/0/2 weight 100

```

```

30 set chassis high-availability services-redundancy-group 1 monitor monitor-object L2-GWS
   interface interface-name ge-0/0/1 weight 100
31 set chassis high-availability services-redundancy-group 1 monitor srg-threshold 100
32 set chassis high-availability services-redundancy-group 1 active-signal-route 169.254.200.1
33 set chassis high-availability services-redundancy-group 1 backup-signal-route 169.254.200.2
34 set chassis high-availability services-redundancy-group 1 activeness-priority 100
35 set security policies default-policy permit-all
36 set security zones security-zone HA-ICL host-inbound-traffic system-services all
37 set security zones security-zone HA-ICL host-inbound-traffic protocols all
38 set security zones security-zone HA-ICL interfaces ge-0/0/3.0
39 set security zones security-zone trust host-inbound-traffic system-services all
40 set security zones security-zone trust host-inbound-traffic protocols all
41 set security zones security-zone trust interfaces ge-0/0/1.0
42 set security zones security-zone trust interfaces ge-0/0/0.500
43 set security zones security-zone trust interfaces ge-0/0/0.10
44 set security zones security-zone untrust host-inbound-traffic system-services ping
45 set security zones security-zone untrust interfaces ge-0/0/2.0
46 set interfaces ge-0/0/0 flexible-vlan-tagging
47 set interfaces ge-0/0/0 native-vlan-id 10
48 set interfaces ge-0/0/0 unit 10 vlan-id 10
49 set interfaces ge-0/0/0 unit 10 family inet address 10.10.99.195/26
50 set interfaces ge-0/0/0 unit 500 vlan-id 500
51 set interfaces ge-0/0/0 unit 500 family inet address 192.168.253.3/24
52 set interfaces ge-0/0/1 unit 0 family inet address 10.10.99.131/26
53 set interfaces ge-0/0/2 unit 0 family inet address 192.168.252.253/24
54 set interfaces ge-0/0/3 mtu 1514
55 set interfaces ge-0/0/3 unit 0 family inet address 100.0.0.1/31
56 set interfaces fxp0 unit 0 family inet address 192.168.100.181/24
57 set interfaces lo0 unit 0 family inet address 172.16.0.2/32
58 set forwarding-options hash-key family inet layer-3
59 set forwarding-options hash-key family inet layer-4
60 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from protocol direct
61 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from route-filter 192.168.252.0/24
   exact
62 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 from condition
   ACTIVE_ROUTE_EXISTS_SRG1
63 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 then metric 10
64 set policy-options policy-statement MNHA_ROUTE_POLICY term 1 then accept
65 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from protocol direct
66 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from route-filter 192.168.252.0/24
   exact
67 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 from condition
   BACKUP_ROUTE_EXISTS_SRG1
68 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 then metric 20
69 set policy-options policy-statement MNHA_ROUTE_POLICY term 2 then accept
70 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 from protocol direct
71 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 then metric 30
72 set policy-options policy-statement MNHA_ROUTE_POLICY term 99 then accept
73 set policy-options policy-statement MNHA_ROUTE_POLICY term default then reject
74 set policy-options policy-statement load-balancing-policy then load-balance per-flow
75 set policy-options condition ACTIVE_ROUTE_EXISTS_SRG1 if-route-exists address-family inet
   169.254.200.1/32
76 set policy-options condition ACTIVE_ROUTE_EXISTS_SRG1 if-route-exists address-family inet table
   inet.0
77 set policy-options condition BACKUP_ROUTE_EXISTS_SRG1 if-route-exists address-family inet
   169.254.200.2/32
78 set policy-options condition BACKUP_ROUTE_EXISTS_SRG1 if-route-exists address-family inet table
   inet.0
79 set protocols bgp group trust type external
80 set protocols bgp group trust export MNHA_ROUTE_POLICY
81 set protocols bgp group trust local-as 65022
82 set protocols bgp group trust multipath multiple-as
83 set protocols bgp group trust bfd-liveness-detection minimum-interval 500
84 set protocols bgp group trust bfd-liveness-detection multiplier 3
85 set protocols bgp group trust bfd-liveness-detection holddown-interval 10000
86 set protocols bgp group trust neighbor 10.10.99.129 peer-as 65012
87 set protocols bgp group trust neighbor 10.10.99.193 peer-as 65012
88 set routing-options static route 0.0.0.0/0 next-hop 192.168.99.1
89 set routing-options forwarding-table export load-balancing-policy

```

1.8 Summary

With any high-availability design prudent planning is imperative. Understanding the requirements, goals, capabilities and integration with the network provides for a successful MNHA deployment:

- Which MNHA model to deploy Layer 3, Layer 2 (Default Gateway) or Hybrid?
- Active/Active or Active/Backup?
- If L3, what protocols will be used to integrate with the network (eBGP, iBGP, OSPF)?
- If L2, what type of fabric to integrate? Spanning Tree concerns?
- Any convergence issues?
- How quick to failover? Automatic Failback?
- What mechanism to track for failover (BFD, IP, Interface, or combination of the 3)?

Once decided, it's just a matter of testing and configuring your deployment.

1.9 Useful links

- Multi-Node High Availability Basics <https://juniper.github.io/techposts/multi-node-high-availability-basics/article>
- Multi-Node High Availability: public documentation <https://www.juniper.net/documentation/us/en/software/junos/high-availability/topics/topic-map/mnha-introduction.html>
- BFD Liveness Detection (BGP) <https://www.juniper.net/documentation/us/en/software/junos/cli-reference/topics/ref/statement/bfd-liveness-detection-edit-protocols-bgp.html>
- Understanding How BFD Detects Network Failures <https://www.juniper.net/documentation/us/en/software/junos/high-availability/topics/topic-map/bfd.html>
- Load Balancing on Aggregated Ethernet Interfaces <https://www.juniper.net/documentation/us/en/software/junos/high-availability/topics/topic-map/load-balancing-aggregated-ethernet-interfaces.html>

1.10 Glossary

- AS: Autonomous System
- BFD: Bi-Directional Forwarding Detection
- BGP: Border Gateway Protocol
- HA: High Availability
- ICD: Inter Chassis Datapath Link
- ICL: Inter-Chassis Link
- L2: Layer 2
- L3: Layer 3
- MED: Multi-Exit Discriminator
- SRG: Services Redundancy Group
- VIP: Virtual IP

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